

hours battery life, typically, it uses AS&E's signature Z Backscatter technology, producing a real-time image of the scan target, bringing organic materials that transmission X-ray systems can miss, such as drug, currency and explosives to the fore by wirelessly connecting to the tablet. Unlike portable transmission X-ray systems, trace detectors or density meters, the MINI Z system provides an easy-to-interpret image to swiftly pinpoint organic contraband behind non-metallic surfaces. The operator controls the scanning function with the help of an uncomplicated, intuitive graphical user interface on the system's dedicated tablet which produces an image of the scanned area that appears in real time.

The MINI Z system is safe for bystanders, operators and the environment as well. The system's X-ray dose conforms to the appropriate EURATOM, NCRP, ICRP and ANSI radiation safety standards. It cannot be used for scanning people, although the MINI Z is a low-dose and low-energy Z Backscatter system.

Some of the applications it can be used for are public safety (quickly screening unattended and suspicious bags and packages for potential terrorist threats), border and security checkpoints (screening vehicle bumpers, tires, panels or interiors for concealed threats such as drugs, cigarettes, explosives and plastic weapons), drug enforcement (investigation of suspected drug labs or fake currency), event security (screening of hand baggage and deliveries to ensure public safety) and VIP security (securing rooms and furniture for organic threats and IEDs to ensure personnel safety).

Wearable electronics to enforce law and order

A body-worn video camera for law enforcement best fits police, private security agencies and other law enforcement personnel. Seattle-based VIEU recently announced its latest offering in the wearable camera domain—the LE3. It is specifically designed for professionals who need to video docu-



Bosch's DINION 8000 (Courtesy: www.us.boschsecurity.com)



Comparison of images from Bosch DINION IP using starlight technology and other cameras (Courtesy: www.us.boschsecurity.com)

ment their actions. In the US, notably the police departments from Fairfield and Novato have deployed these cameras. While some police officers feel that this turns them into RoboCops, others feel more positively and consider it to be a means to prove their side of the story. Despite some objections raised, like privacy issues and mixed reactions from the officers, the use of this technology is growing.

This compact, 70gm, pager-sized, self-contained video recorder with a 16GB non-removable storage captures and provides convincing forensic video evidence, and can be comfortably clipped on the uniform. It has the most uncomplicated way of operation; it turns on/off by simply sliding the lens slide cover, which is very convenient for activation in stressful situations. With two video resolution modes (1280×720 (HD) or 848×480 (widescreen SD)), and battery life up to five hours for SD and three hours for HD, it records at 30 frames per second for up to 12 hours at SD resolution and up to 6 hours at HD. Video evidence is securely stored and tabulated with a FIPS 140-2 compliant digital signature process which can prove that the video has not been tampered with.

The LE3 camera is powered by VIEU's VERIPATRO software system and can be integrated into existing video systems or other video manage-

ment platforms. All video evidence can be accessed and stored locally or in a cloud for ongoing cases and trials. The software installs either in a stand-alone, network or cloud-based mode.

The ability to function in both high-definition and standard-definition modes was one of the most notable features incorporated. This was achieved by adopting the H.264 compression scheme and by increasing the camera's internal memory from 4 GB to 16 GB. Another notable change from its previous version was the change in the lens itself. After consulting with a forensic expert, the company settled on the concept of a flat image for the camera, which means no distortion because of the field of view. The ideal field of view is 65 to 70 degrees, and hence the lens was designed with a 68-degree field of view. That provides a very flat image with no curvature on the back side of the image. So, if one needs to forensically enhance something, it is of high quality all the way through.

Some other improvements designed into this camera, in contrast with the previous LE2, were ability to mute audio. It was incorporated considering the laws in some states that require two-party consent to record conversations and an improved battery life. It costs about US\$ 900.

Smarter surveillance with advanced image processing

Taking into account the burgeoning requirement for security and safety products, more and more cameras are being stationed in neighbourhoods, districts or other locales where they are at the mercy of nature and other elements. Capturing video in dim and dingy areas, where most of the cameras fail, and providing colour images when cameras transition to monochrome mode, is a challenge overcome by this 5MP camera with starlight technology. Even at night, the DINION IP starlight 8000 MP benefits users by providing details